

| ID   | Module       | Test Case                     | Expected Result                                   | Actual Result           | Status |
|------|--------------|-------------------------------|---------------------------------------------------|-------------------------|--------|
| TC18 | Payment      | Razorpay checkout             | Payment processed, status updated                 | Payment successful      | Pass   |
| TC19 | Carbon       | Calculate carbon offset       | Correct CO <sub>2</sub> value based on waste type | Values match factors    | Pass   |
| TC20 | Security     | Unauthenticated access        | Redirected to login page                          | Access denied           | Pass   |
| TC21 | Location     | GPS coordinate capture        | Latitude/longitude stored with listing            | Coordinates saved       | Pass   |
| TC22 | Location     | Reverse geocoding             | Address string returned from coordinates          | Address displayed       | Pass   |
| TC23 | Notification | Status change notification    | Notification created in Firestore                 | Notification sent       | Pass   |
| TC24 | Notification | Unread count badge            | Badge shows correct unread count                  | Count shown correctly   | Pass   |
| TC25 | Farmer       | Export CSV                    | CSV file downloaded with listing data             | CSV exported            | Pass   |
| TC26 | Farmer       | View carbon credits           | Carbon panel shows offset values                  | Credits displayed       | Pass   |
| TC27 | Agent        | Claim already-claimed listing | Error message shown                               | Error message displayed | Pass   |
| TC28 | Admin        | View platform statistics      | User and listing counts displayed                 | Statistics shown        | Pass   |
| TC29 | Responsive   | Mobile viewport rendering     | All elements visible and usable                   | Layout responsive       | Pass   |
| TC30 | Performance  | Dashboard load time           | Page loads within 3 seconds                       | Loaded in 1.8s          | Pass   |

Table 10.1: Test Cases and Results

### 10.3 Performance Benchmarks

The following table presents performance benchmarks measured during testing: